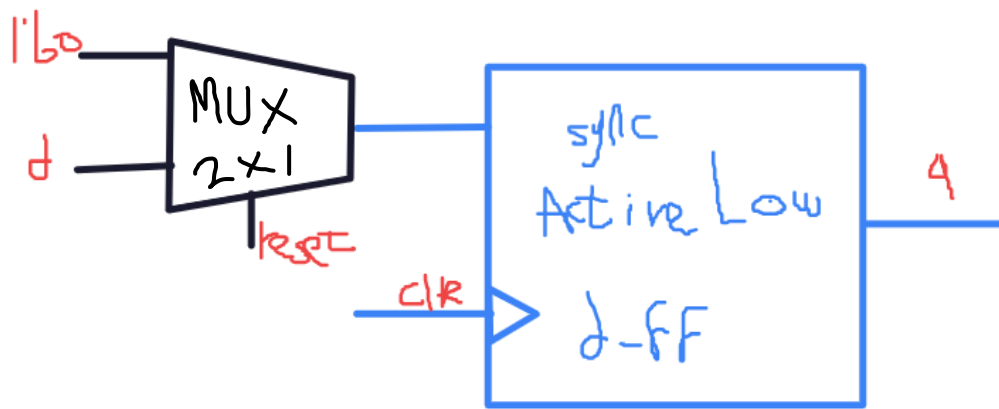




```

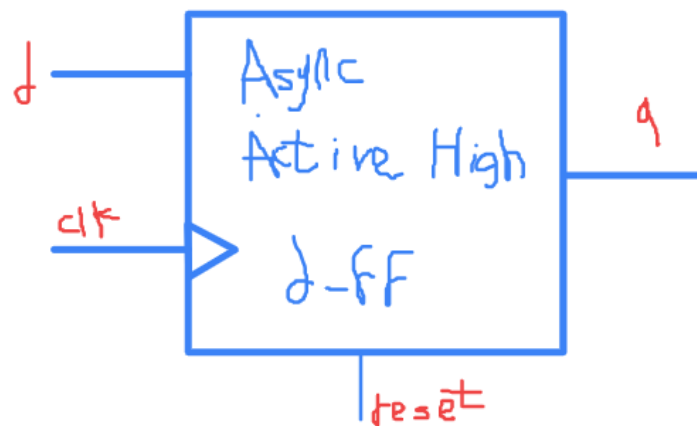
module d_ff_sync(
    input wire d,
    input wire clk,
    input wire reset, //active low & synchronous reset

    output reg q
);

always@(posedge clk)
begin
    if(!reset) q<=0'b0;
    else q<=d;
end

endmodule

```



```

module d_ff_async(
    input wire d,
    input wire clk,
    input wire reset, //active high & Asynchronous reset

    output reg q
);

always@(posedge clk or posedge reset)
begin
    if(reset) q<=0'b0;
    else q<=d;
end

endmodule

```